THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 5: **Measurement of Length and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The thickness of a hair is about 0.1 mm. In metres it is: 1

(a) 0.0001 m **(b)** 0.01 m **(c)** 0.1 m **(d)** 0.001 m

2. Which measurement is most accurate? 1

(a) 12.4 cm **(b)** 12.35 cm **(c)** 12 cm **(d)** 12.3 cm

3. The Earth rotates once on its axis in about: 1

(a) 12 hours **(b)** 365 days **(c)** 27 days **(d)** 24 hours

4. A measuring tape has marks in cm on one side and inches on the other. This is: 1

(a) an error **(b)** useless

(c) for using two unit systems **(d)** for adding

5. If a ruler is calibrated in mm, the least count is: 1

(a) 1 mm **(b)** 0.1 mm **(c)** 1 cm **(d)** 1 m

6. The Earth revolves around the Sun in about: 1

(a) 30 days **(b)** 27 days **(c)** 365 days **(d)** 24 hours

7. Walking 5 steps each of 0.6 m, the distance walked is: 1

(a) 0.6 m **(b)** 5 m **(c)** 3 m **(d)** 30 m

8. A metre scale has the zero mark worn. A reading from 1 cm to 11.6 cm means length: 1

(a) 11.6 cm **(b)** 10.6 cm **(c)** 12.6 cm **(d)** 1 cm

9. Which of these shows that motion is relative? 1

(a) a stone sinks

(b) a ball falls down

(c) a passenger seems at rest in a moving train

(d) a bird sings

10. Which of the following measurements shows more precision? 1

(a) 5.00 cm **(b)** 5 cm **(c)** about 5 cm **(d)** 5.0 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A scooter wheel of circumference 1.5 m turns 200 times. What distance is covered? 2

12. Give an example where an object is both at rest and in motion at the same time. 2

13. How would you measure the length of a river on a school map? 2

14. Why does the Sun appear to move from east to west? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A runner covers 400 m in each lap. How many laps for 5 km? 3

16. Describe how to measure the width of a river without crossing it using ideas of measurement (any simple method). 3

17. A path has three straight sections of 2.5 m, 450 cm and 0.0036 km. Find total length in metres. 3

OR

Two students measure the length of the same desk and get 1.28 m and 128.4 cm. Explain whether their results agree. 3

18. Explain why the estimate of length by eye is less reliable than measurement. 3

OR

Why does a spring bounce periodically? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the various kinds of motion shown by a bicycle journey (wheels, pedals, handlebar, the cyclist) and classify each. 5

OR

Discuss how the Earth's rotation and revolution lead to day-night and years, linking them with periodic motion. 5

20. Explain with examples how the reference point decides whether an object is at rest or moving. 5

OR

A girl walks around a square park of side 120 m twice. Find the total distance and the displacement at the end (ending at the starting point). 5

END OF QUESTION PAPER — Total: 40 Marks

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